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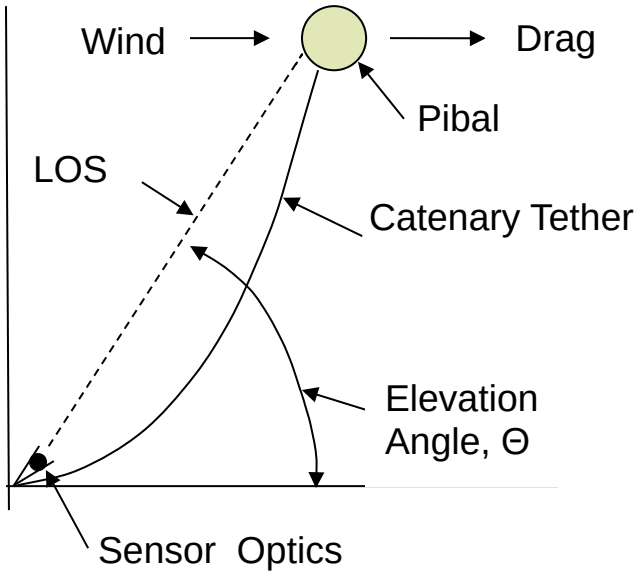
# Tethered Pilot Balloon Bias Errors(rev.1) 27 February 2013

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## Introduction

The continued use of tethered pilot balloons (Pibals) to measure vector winds in the lower parts of the planetary boundary layer provides the motivation for writing this memo. It is customary to assume the tether can be modeled as a perfectly straight string. However, this would violate the laws of physics were it so. This memo documents a study of the errors in the straight tether model due to gravity and aerodynamic drag.

Now consider the geometry of this wind measurement system as described in Ref. (1):




The sketch on the right shows the wind sensor. Note the elevation angle, Θ, measured by an optical sensor where the tether is grounded. As shown in the sketch, the measured elevation angle is not the same as the elevation angle where the tether attaches to the pibal. This difference is due to gravity and aerodynamic drag.

The classical relation between wind speed $V$ and elevation angle $\Theta$ at the top of the tether is found from statics to be

$$V = \sqrt{\frac{2(B - W)\cot\Theta}{\rho S C_D}}\text{, where} \qquad (1)$$

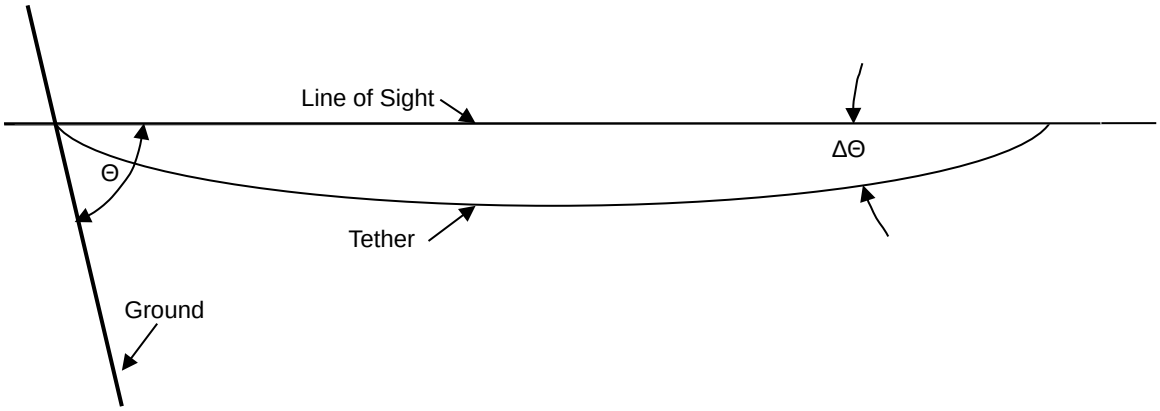
$B$ = Balloon buoyancy force (from Archimedes' Law),
$W$ = Weight of balloon, inflation gas and tether,
$\rho$ = Atmospheric mass density,
$S$ = Balloon projected area = $\pi R^2$, and
$C_D$ = Balloon coefficient of drag.

Strictly speaking the elevation angle appearing in eq. (1) is not that measured on the ground, but instead is that where the tether attaches to the pibal. Our error analysis consists of relating the elevation angle at pibal attachment to that measured on the ground.

Since the effects of gravity and drag are each expected to be small, they are analyzed separately and added to estimate the bias error in eq. (1) due to the analytical approximations in eq. (1).

Gravity

Fortunately, there is a well known closed form solution for the response of a flexible cable to gravity. See Ref. (2) for a brief overview.




As the sketch above shows gravity causes the tether to sag a small amount with respect the line of sight. Noting that the tether weight normal to the line of sight is $W_T \cos\Theta$ ,Ref. (2) gives the condition of static equilibrium at either end to be:

$$\Delta\Theta = \frac{W_T \cos\Theta}{2T}, \text{ where}$$

$W_T$ = Weight of the deployed tether, and
$T$ = Tension force in the tether.

To first order the tension force is

$$T = (B - W)\csc\Theta .$$

Substituting, we have

$$\Delta\Theta = \frac{W_T \sin 2\Theta}{4(B - W)} \qquad (2)$$

Aerodynamic Drag

Estimation of the bias error due to aerodynamic drag follows similar lines. According to Ref. (3) the cross flow drag is effectively decoupled from the longitudinal drag force. Assuming the wind lies in a plane parallel to the ground, the wind normal to the tether is $V \sin\Theta$. An element of tether then experiences a drag load of

$$dD = C_d \frac{1}{2} \rho * 2r * dx V^2 \sin^2\Theta, \text{ where}$$

$C_d$ = 2D cylinder drag coefficient,
$D$ = Transverse drag force acting on the tether,
$r$ = Radius of cylindrical tether, and
$x$ = Distance along the tether.

The result clearly depends on the velocity profile, i.e., how wind speed varies with altitude. If it remains constant with altitude,

$$D = C_d \rho r d V^2 \sin^2\Theta, \text{ where} \tag{3}$$

$d$ = Length of deployed tether.

Then, following the same procedure used to estimate $\Delta\Theta$ for tether weight, we find that

$$\Delta\Theta = \frac{C_d \rho r d V^2 \sin^3\Theta}{2(B - W)} \tag{4}$$

Composite Bias Error

Both bias errors act to increase the elevation angle over its simple line of sight measured value. That is, if $\Theta$ is interpreted as the measured value on the ground, the cotangent appearing in eq. (1) should be replaced with

$$\cot(\Theta + \frac{W_T \sin 2\Theta}{4(B - W)} + \frac{C_d \rho r d V^2 \sin^3\Theta}{2(B - W)}) \equiv \cot\Theta - \csc^2\Theta(\frac{W_T \sin 2\Theta + 2C_d \rho r d V^2 \sin^3\Theta}{4(B - W)}), \text{o}$$

$$\text{r}$$

$$\equiv \cot\Theta\left[1 - \frac{W_T + C_d \rho r d V^2 \tan\Theta \sin\Theta}{2(B - W)}\right] \tag{5}$$

Now, consider numbers for a typical case. Let

$d$ = 600 ft,
$h$ = 500 ft ,
$\Theta = \sin^{-1}\frac{500}{600} = 0.985111 rad = 56.44^\circ$ ,and
$R = 1.5$ ft ==>Balloon diameter = 3 ft,

S = 7.07 $ft^2$,
Vol = 14.137 $ft^3$,
$C_D = 0.49$,
$\rho = 0.002378$ sl/$ft^3$ (sea level standard day)
$\rho_{He} = 0.002378 * 4.02 / 28.966 = 0.000330$ sl/$ft^3$ (He at sea level)
$g = 32.174$ ft/$s^2$,
$B = 1.082$ lb (from Archimedes' law),
$W = 100gr = 0.22$ lb (balloon empty weight)
$W_{He} = 0.15$ lb
$W_{Bal} = 0.22 + 0.15 = 0.37$ lb (total weight of inflated balloon)
$V = 10.7079$ ft/s (from eq.(1))
$V_N = V \sin\Theta = 8.9233$ ft/s normal to the tether
$W_T = 0.024$ lb

$$\frac{W_T}{2(B - W)} = 0.01685$$

$r = 0.0055$ in.
The specific gravity of Spectra™2000 is 0.97 = $4 \times 10^{-5}$ lb/ft. Then
The Reynolds number for cross flow around the tether is

$$\mathrm{Re} = \frac{\rho V(2r)}{\mu}$$

$\mathrm{Re} = 56$ (based on tether diameter at sea level)
$C_d = 1.92$ at this Re

$$\frac{C_d \rho r d V^2 \tan\Theta \sin\Theta}{2(B - W)} = 0.08820$$

The total correction factor in eq.(5) = 1-0.01685-0.08820 = 0.89495.
Hence the bias errors act to overestimate the wind speed by ~5.4%

Recommendation

Since the bias errors turn out to be fairly significant, the recommended data processing procedure is to first compute wind speed from eq. (1), and then to compute a bias correction factor from eq. (5) using the approach outlined above. Use this to obtain an improved wind speed estimate from eq. (1).

References


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